EX 1: SIMULATION OF FDMA IN MATLAB

Aim: To simulate a Frequency Division Multiple Access (FDMA) system using MATLAB and analyze its communication performance in terms of channel bandwidth, data rate, and signal-to-noise ratio (SNR).

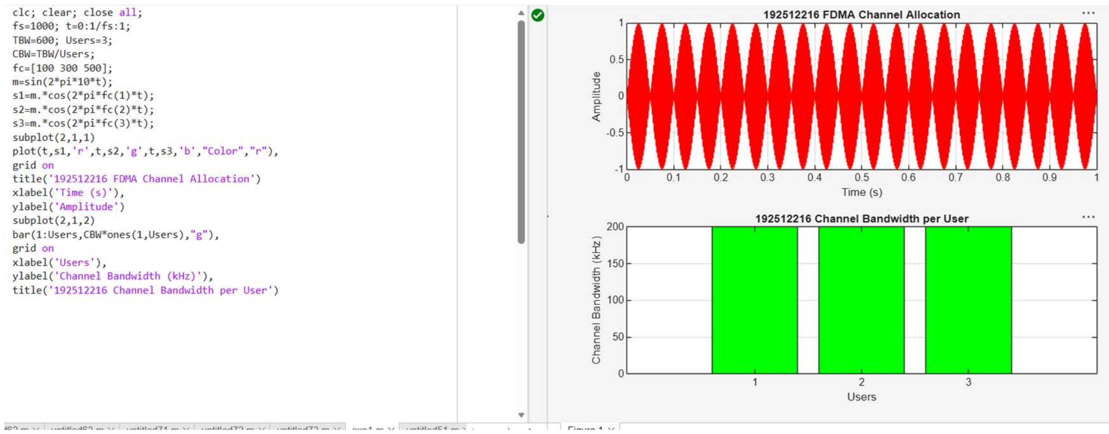
Objective 1: Program

(Matlab Code)

```
clc; clear; close all; fs=1000;
t=0:1/fs:1;
TBW=600; Users=3; CBW=TBW/Users;
fc=[100 300 500]; m=sin(2*pi*10*t);
s1=m.*cos(2*pi*fc(1)*t);
s2=m.*cos(2*pi*fc(2)*t);
s3=m.*cos(2*pi*fc(3)*t);
subplot(2,1,1)
plot(t,s1,'r',t,s2,'g',t,s3,'b'),grid on
title('FDMA Channel Allocation'),xlabel('Time
(s)'),ylabel('Amplitude') subplot(2,1,2)
bar(1:Users,CBW*ones(1,Users)),grid on
xlabel('Users'),ylabel('Channel Bandwidth (kHz)'),title('Channel
Bandwidth per User')
```

Output:

Objective 1:



Output Graph 1: FDMA Channel Allocation

Objective 2: Matlab Code and Output

clc; clear; close all;

BW = 200; % Channel Bandwidth (kHz)

Users = 3;

M = 2; % BPSK Modulation

DataRate = BW*log2(M); % Data Rate (kbps)

Rate = DataRate*ones(1,Users); disp(' User

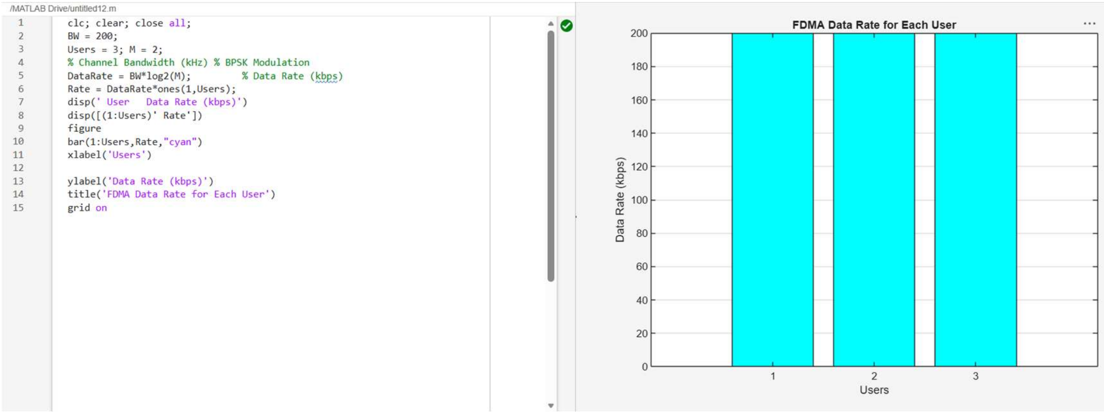
Data Rate (kbps)') disp([(1:Users)' Rate'])

figure bar(1:Users,Rate) xlabel('Users')

ylabel('Data Rate (kbps)')

title('FDMA Data Rate for Each User') grid

on



Objective 3: Matlab Code and Output

```
clc; clear; close all;
Ps = 1; % Signal Power (W)
Pn = [0.1 0.05 0.01]; % Noise Power (W)
Users = 1:3;
SNR = 10*log10(Ps./Pn); % SNR in dB
disp('User Noise Power(W) SNR(dB)')
disp([Users' Pn' SNR']) figure
bar(Users,SNR) xlabel('Users')
ylabel('SNR (dB)')
title('Signal-to-Noise Ratio of FDMA Users') grid
on
```

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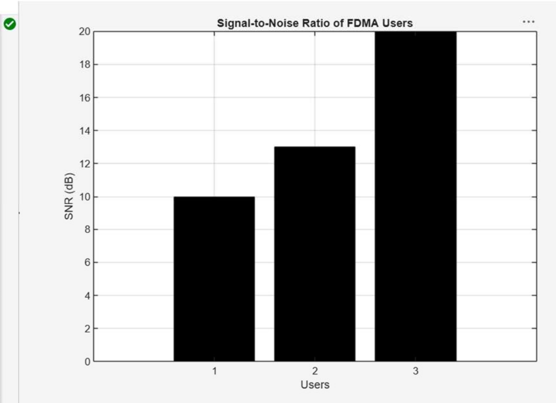
```matlab
clc; clear; close all;
Ps = 1;
Pn = [0.1 0.05 0.01];            % Signal Power (W) % Noise Power (W)
Users = 1:3; SNR = 10*log10(Ps./Pn);          % SNR in dB
disp('User   Noise Power(W)   SNR(dB)')
disp([Users' Pn' SNR'])
figure
bar(Users,SNR,"black")
xlabel('Users')
ylabel('SNR (dB)')
title('Signal-to-Noise Ratio of FDMA Users')
grid on
```



Result:

1. The FDMA system was successfully simulated by allocating separate frequency channels to three users, and the channel bandwidth assigned to each user was determined.

2. The data rate achieved by each user was successfully calculated based on the allocated channel bandwidth, demonstrating equal transmission capacity for all users.

3. The Signal-to-Noise Ratio (SNR) for different users was successfully evaluated, showing that higher SNR values provide better communication quality and reliable data transmission.